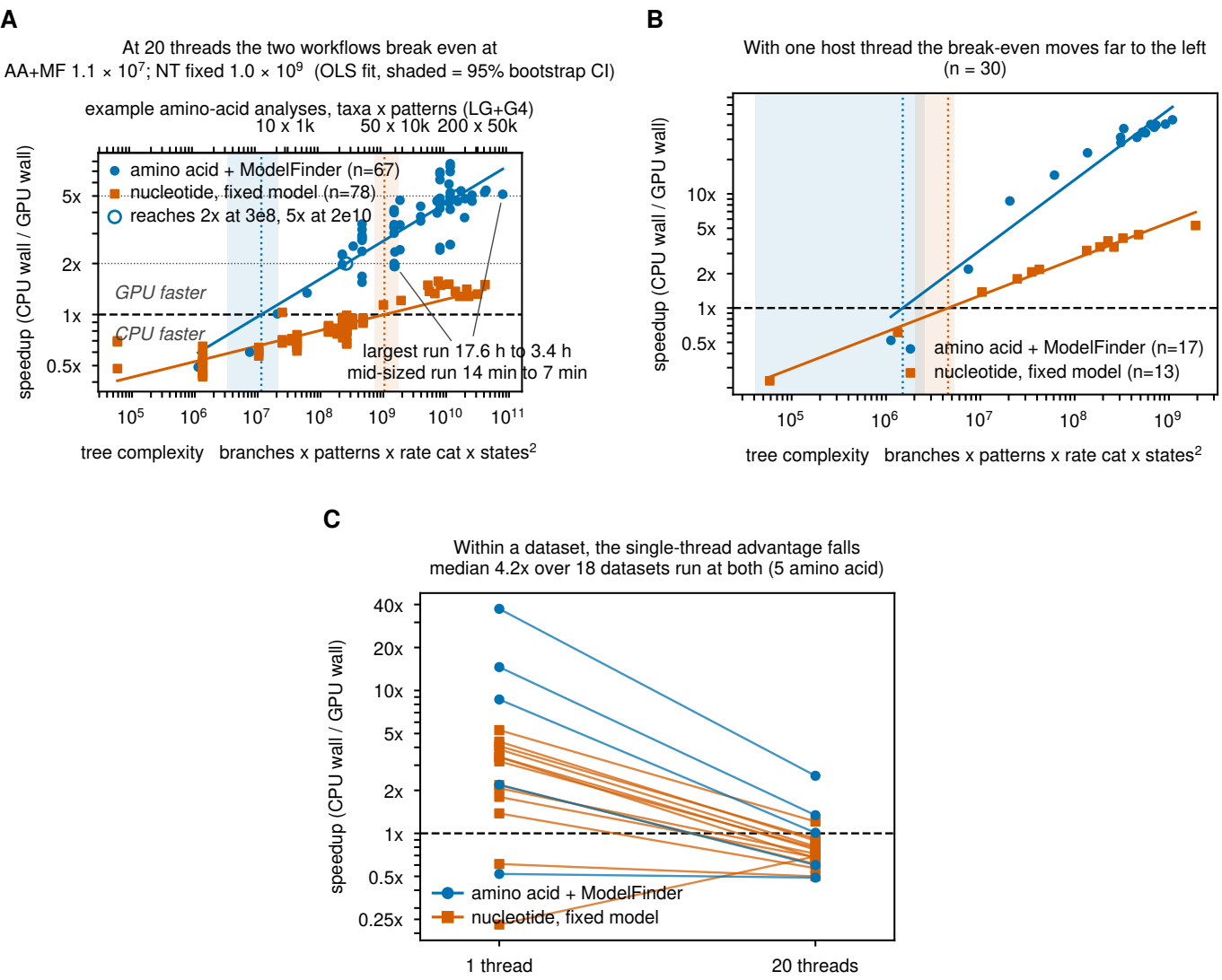

A
At 20 threads the two workflows break even at
AA+MF 1.1 × 10^7; NT fixed 1.0 × 10^9 (OLS fit, shaded = 95% bootstrap CI)
example amino-acid analyses, taxa x patterns (LG+G4)
10 x 1k
50 x 10k
200 x 50k
amino acid + ModelFinder (n=67)
nucleotide, fixed model (n=78)
reaches 2x at 3e8, 5x at 2e10
GPU faster
CPU faster
largest run 17.6 h to 3.4 h
mid-sized run 14 min to 7 min
speedup (CPU wall / GPU wall)
5x
2x
1x
0.5x
10^5
10^6
10^7
10^8
10^9
10^10
10^11
tree complexity branches x patterns x rate cat x states^2
B
With one host thread the break-even moves far to the left
(n = 30)
speedup (CPU wall / GPU wall)
10x
5x
2x
1x
0.5x
amino acid + ModelFinder (n=17)
nucleotide, fixed model (n=13)
10^5
10^6
10^7
10^8
10^9
tree complexity branches x patterns x rate cat x states^2
C
Within a dataset, the single-thread advantage falls
median 4.2x over 18 datasets run at both (5 amino acid)
speedup (CPU wall / GPU wall)
40x
20x
10x
5x
2x
1x
0.5x
0.25x
amino acid + ModelFinder
nucleotide, fixed model
1 thread
20 threads